Annex V. Liberia: Debt Sustainability Analysis[1]

This Debt Sustainability Analysis (DSA) updates the analysis presented to the Board in December 2015. Liberia's risk of debt distress remains moderate but debt vulnerabilities have risen. The debt-to-exports and debt-to-GDP ratios have deteriorated compared to the last review, reflecting the worsened exports and growth outlook.[2] The economy remains vulnerable to negative shocks to terms of trade, borrowing terms, exports, and growth.

A. Underlying Assumptions

1. The baseline macroeconomic projections in the near to medium term have deteriorated from the December 2015 DSA (Text table 1). Reflecting the larger than expected adverse impact of the commodity price shock, GDP growth has been revised down to an average of 3.4 percent during 2015–20, down from 4.1 percent in the fourth review staff report, while exports growth is also expected to be slower than previously projected. Over the longer term, macroeconomic projections remain broadly unchanged. Despite weakened outlook, external debt significantly accelerated in recent years, rising from 11 percent of GDP at end-2013 to 23 percent of GDP at end-2015, and is expected to increase further over the medium term.

Text Table 1. Underlying DSA assumptions
(percent of GDP unless otherwise indicated)

	2015-2020		2021-2035	
	Current DSA	Previous DSA	Current DSA	Previous DSA
Real GDP growth (in percent)	3.4	4.1	6.1	6.1
GDP Deflator (percent change)	2.9	2.2	2.3	2.5
Primary deficit	5.9	5.9	0.8	0.8
Exports growth (in percent)	-4.0	-3.7	8.6	9.4

Sources: The Liberian authorities; and IMF staff projections.

B. External DSA

2. The external DSA indicates a moderate risk of debt distress in the baseline scenario. In the medium term, the PV of debt-to-GDP ratio is projected to rise steadily from 14 percent in FY2015 to 22 percent by FY2020 in the baseline scenario, and gradually decline thereafter. The deteriorated exports outlook, mainly owing to the commodity price shock, has contributed to the projected increase in the PV of debt-to-export ratio from 37 percent in FY2015 to a peak of 96 percent in FY2021, close to the 100 percent baseline threshold. The evolution of debt-exports ratio suggests a "moderate-to-high borderline case", while the probability charts do not show any breach of thresholds in the baseline, confirming a moderate risk of debt distress (Figure 3). Due to the highly concessional nature of new debt, debt-to revenues and debt service indicators remain well below the thresholds.

3. However, debt vulnerabilities have risen, with worsened PV of debt-to-GDP and debt-to-exports ratios under alternative scenarios. Under the extreme shock scenario of adverse terms of trade shocks, both the PV of debt-to-exports and external debt-to-GDP ratios breach the threshold around FY 2019–20 and remain higher than the threshold for an extended period. Both the

[1] Prepared by Cindy Xu.

[2] The risk of debt distress is based on country-specific policy-dependent thresholds, with Liberia's CPIA index at 3.09, which is in the category of weak policy performance.

magnitude and persistence of the breach have increased compared to the previous DSA, indicating intensified debt vulnerabilities. Further, the current risk rating is very sensitive to export dynamics, with any further deterioration in exports would likely push the debt rating from moderate to high. As in the previous DSA, the PV of debt to revenue and debt service related ratios remain well below thresholds in all scenarios. The authorities agreed with staff's assessment of moderate risk rating and shared staff's concerns on debt vulnerabilities. They plan to streamline new borrowing over the medium term focusing only on existing pipeline projects while continuing to prioritize grants and concessional financing. Nevertheless, they highlighted the tension arising from debt sustainability and the need to close infrastructure gaps, and noted that the growth dividends would help debt sustainability. The authorities also called for more donor coordination as some of the externally-financed projects were not high priority but would further increase the debt burden.

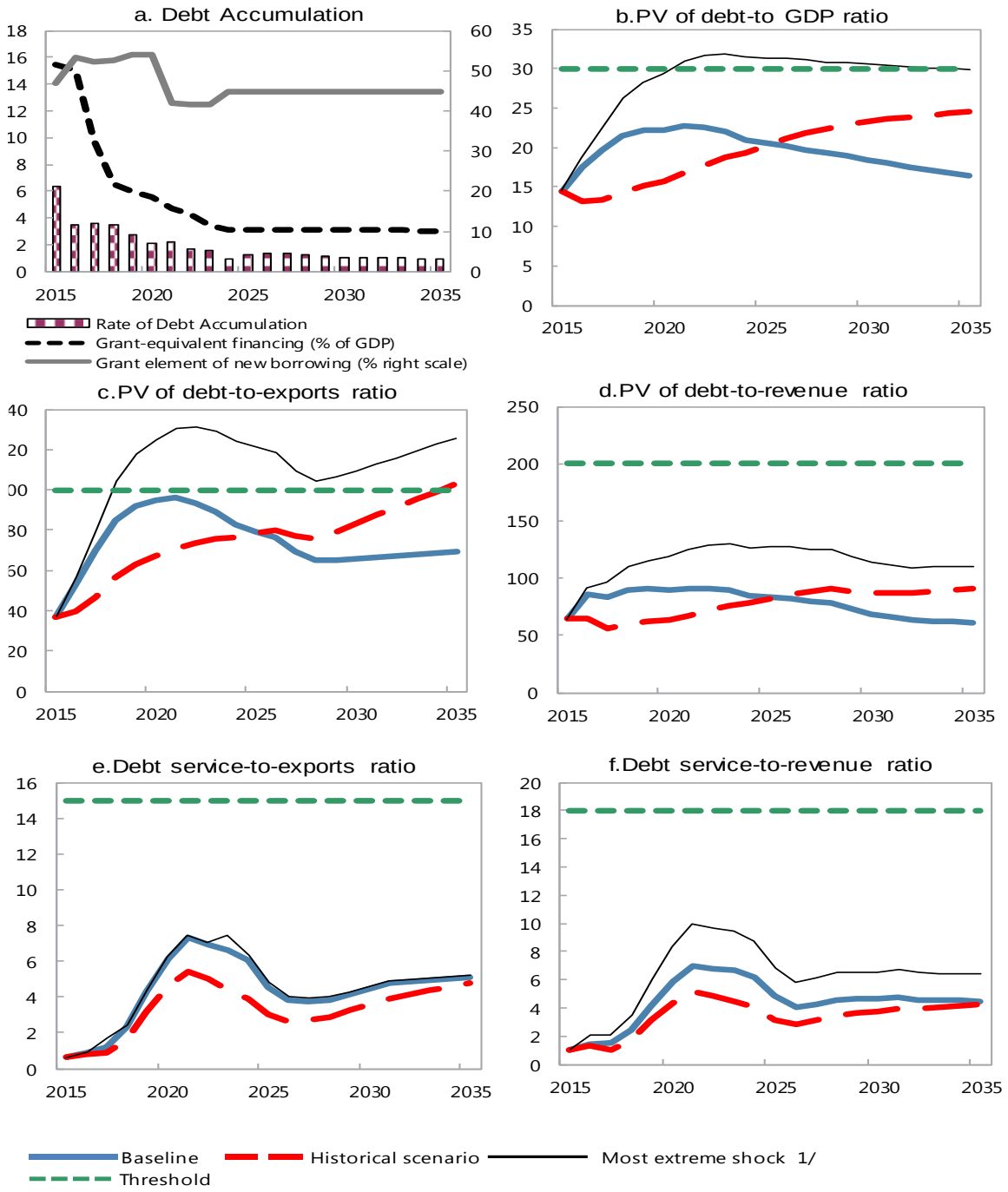
C. Public DSA

4. The public DSA has not significantly changed compared to the December update. Under the baseline and shock scenarios, all debt indicators remain below their respective thresholds. Under the baseline, the PV of public debt-to-GDP rises steadily from 16 percent in FY2015 to around 22 percent in FY2020 and is expected to fall afterwards. The PV of debt-to-revenue ratio rises to a peak of 81 percent in FY2019 and then follows a gradual downward path towards the end of the projection period. The PV of debt service-to-revenue ratio reaches its peak of 7 percent in FY2021 and projected to decline afterwards. Under alternative scenarios, public debt indicators are particularly sensitive to adverse shocks to lower growth, a large real depreciation, and a significant increase in other debt-creating flows.

D. Conclusion

5. Increasing debt vulnerabilities call for prudent borrowing policy and economic diversification, despite the moderate risk rating. The latest DSA results indicate a rise in debt vulnerabilities compared with the previous review, underlined most notably by the deterioration of debt to exports and debt to GDP ratios in the extreme shock scenario, though the risk of debt distress remains moderate as confirmed by the probability approach. Nevertheless, the debt rating is very close to the high risk category and particularly vulnerable to export shocks. In the case of a high risk rating caused by any further deterioration in debt and exports outlook, the debt limit conditionality would be reassessed in line with the new debt limit policy.

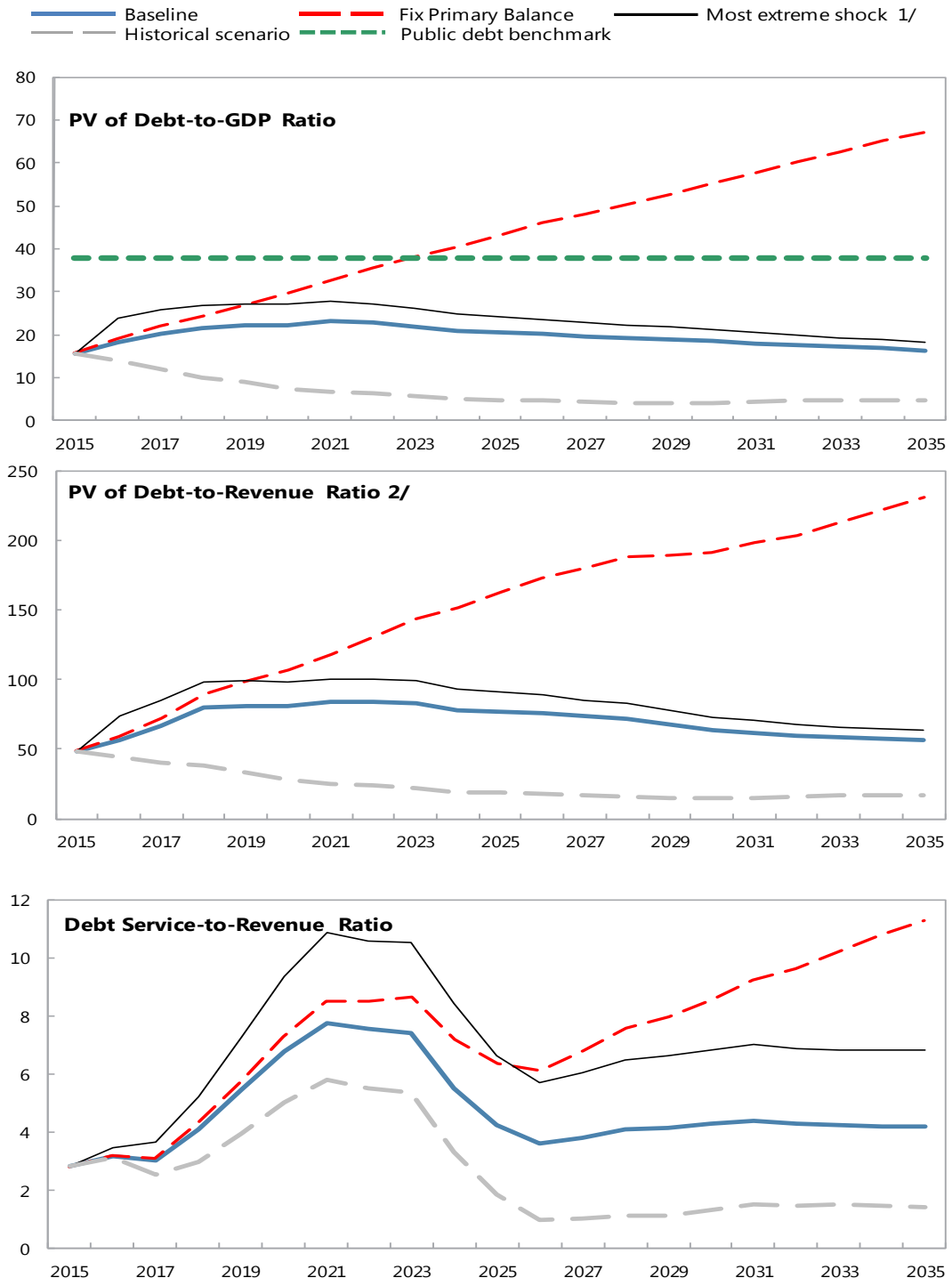
Figure A1. Liberia: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.

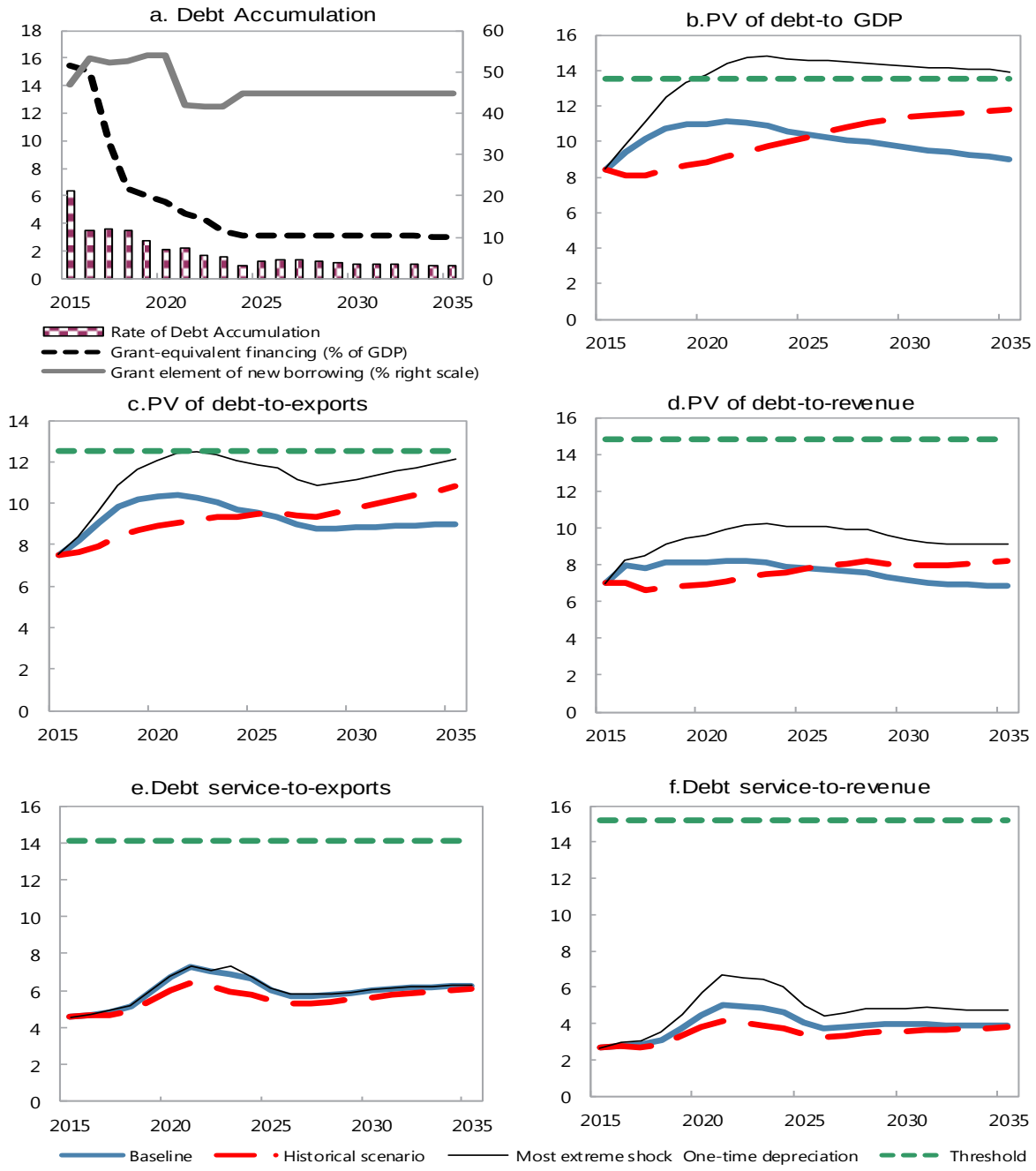
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Terms shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Non-debt flows shock and in figure f. to a One-time depreciation shock.

Figure A2. Liberia: Indicators of Public Debt under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Figure A3. Liberia: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2015–35[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a Terms shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Non-debt flows shock and in figure f. to a One-time depreciation shock.

Table A1. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2012–35[1]

(Percent of GDP, fiscal year, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**9.4**	**10.9**	**13.6**			**23.4**	**29.2**	**33.2**	**36.5**	**38.3**	**39.1**		**35.1**	**27.2**	
of which: public and publicly guaranteed (PPG)	9.4	10.9	13.6			23.4	29.2	33.2	36.5	38.3	39.1		35.1	27.2	
Change in external debt	-2.2	1.5	2.7			9.8	5.8	4.0	3.3	1.8	0.9		-1.2	-0.7	
Identified net debt-creating flows	-0.2	2.6	7.8			12.8	9.5	6.4	4.2	2.9	2.3		-0.6	0.3	
Non-interest current account deficit	**24.3**	**25.0**	**30.2**	**21.4**	**11.7**	**33.5**	**32.3**	**29.0**	**26.9**	**26.3**	**26.3**		**30.3**	**24.2**	28.3
Deficit in balance of goods and services	77.3	65.3	73.9			90.9	86.6	69.4	56.4	54.1	51.2		43.9	33.3	
Exports	48.3	48.4	44.2			39.1	33.0	28.3	25.3	24.0	23.5		25.8	23.8	
Imports	125.6	113.8	118.1			130.0	119.5	97.8	81.7	78.2	74.7		69.7	57.1	
Net current transfers (negative = inflow)	-66.5	-57.4	-61.7	-99.6	38.2	-73.1	-66.7	-51.3	-40.8	-38.2	-35.7		-26.0	-15.9	-23.0
of which: official	-26.7	-25.1	-32.2			-37.6	-31.1	-27.3	-25.4	-23.4	-21.7		-14.6	-8.7	
Other current account flows (negative = net inflow)	13.5	17.1	17.9			15.7	12.4	10.9	11.3	10.4	10.8		12.3	6.8	
Net FDI (negative = inflow)	**-22.9**	**-21.5**	**-21.6**	**-13.9**	**17.0**	**-21.2**	**-22.1**	**-21.8**	**-21.5**	**-21.9**	**-22.2**		**-28.9**	**-23.1**	-26.7
Endogenous debt dynamics 2/	**-1.6**	**-0.9**	**-0.7**			**0.5**	**-0.7**	**-0.8**	**-1.1**	**-1.5**	**-1.7**		**-2.0**	**-0.8**	
Contribution from nominal interest rate	0.0	0.1	0.1			0.1	0.2	0.2	0.3	0.3	0.3		0.4	0.3	
Contribution from real GDP growth	-0.8	-0.7	-0.6			0.4	-0.9	-1.0	-1.4	-1.8	-2.0		-2.4	-1.1	
Contribution from price and exchange rate changes	-0.8	-0.4	-0.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-2.1**	**-1.2**	**-5.1**			**-3.0**	**-3.7**	**-2.4**	**-1.0**	**-1.1**	**-1.4**		**-0.6**	**-1.0**	
of which: exceptional financing	0.0	0.0	0.0			-0.9	-0.9	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	8.0			14.5	17.6	19.8	21.5	22.1	22.2		20.5	16.4	
In percent of exports	...	...	18.2			37.1	53.3	69.7	84.9	92.1	94.5		79.5	69.0	
PV of PPG external debt	**...**	**...**	**8.0**			**14.5**	**17.6**	**19.8**	**21.5**	**22.1**	**22.2**		**20.5**	**16.4**	
In percent of exports	**...**	**...**	**18.2**			**37.1**	**53.3**	**69.7**	**84.9**	**92.1**	**94.5**		**79.5**	**69.0**	
In percent of government revenues	**...**	**...**	**34.2**			**64.5**	**85.9**	**83.7**	**90.0**	**90.3**	**89.9**		**83.5**	**60.6**	
Debt service-to-exports ratio (in percent)	**0.1**	**0.8**	**0.4**			**0.6**	**0.9**	**1.2**	**2.3**	**4.3**	**6.2**		**4.6**	**5.1**	
PPG debt service-to-exports ratio (in percent)	**0.1**	**0.8**	**0.4**			**0.6**	**0.9**	**1.2**	**2.3**	**4.3**	**6.1**		**4.6**	**5.1**	
PPG debt service-to-revenue ratio (in percent)	**0.2**	**1.4**	**0.8**			**1.1**	**1.4**	**1.5**	**2.4**	**4.2**	**5.9**		**4.8**	**4.5**	
Total gross financing need (Millions of U.S. dollars)	24.1	73.4	175.9			250.3	218.4	167.8	142.7	144.3	158.8		111.7	226.2	
Non-interest current account deficit that stabilizes debt ratio	26.5	23.6	27.4			23.7	26.4	25.0	23.6	24.6	25.4		31.4	24.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.9	8.5	5.7	7.1	1.8	-2.8	3.9	3.6	4.5	5.5	5.8	3.4	7.1	4.2	6.1
GDP deflator in US dollar terms (change in percent)	7.6	4.0	2.3	6.3	2.7	2.2	0.8	3.1	3.1	4.3	3.7	2.9	1.4	3.8	2.3
Effective interest rate (percent) 5/	0.1	1.6	0.7	0.4	0.6	0.9	0.8	0.8	0.8	0.8	0.9	0.8	1.2	1.2	1.2
Growth of exports of G&S (US dollar terms, in percent)	26.1	13.1	-1.4	15.3	16.5	-12.1	-11.6	-8.2	-3.9	4.6	7.4	-4.0	10.6	4.7	8.6
Growth of imports of G&S (US dollar terms, in percent)	9.3	2.2	12.2	16.0	23.9	9.4	-3.7	-12.6	-10.0	5.2	4.9	-1.1	4.6	5.0	6.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	46.9	53.2	52.1	52.7	53.9	53.9	52.1	44.6	44.6	44.0
Government revenues (excluding grants, in percent of GDP)	26.2	27.5	23.5			22.5	20.5	23.6	23.8	24.5	24.7		24.6	27.0	25.8
Aid flows (in Millions of US dollars) 7/	28.3	45.7	77.6			199.1	243.6	148.6	77.9	77.7	82.8		87.5	196.7	
of which: Grants	28.3	45.7	77.6			199.1	243.6	148.6	77.9	77.7	82.8		87.5	196.7	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			15.5	15.1	9.8	6.5	6.0	5.6		3.1	3.0	3.3
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			71.4	83.5	77.1	69.0	69.7	70.4		69.2	70.3	68.3
Memorandum items:															
Nominal GDP (Millions of US dollars)	1643.1	1853.7	2004.7			1991.4	2085.6	2228.6	2400.9	2639.8	2896.3		4372.7	9836.6	
Nominal dollar GDP growth	16.0	12.8	8.1			-0.7	4.7	6.9	7.7	10.0	9.7	6.4	8.6	8.2	8.5
PV of PPG external debt (in Millions of US dollars)			156.5			283.4	353.5	428.5	506.2	572.8	628.0		882.6	1588.1	
(PVt-PVt-1)/GDPt-1 (in percent)						**6.3**	3.5	3.6	3.5	2.8	2.1	3.6	1.3	0.9	1.3
Gross workers' remittances (Millions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	8.0			14.5	17.6	19.8	21.5	22.1	22.2		20.5	16.4	
PV of PPG external debt (in percent of exports + remittances)	...	...	18.2			37.1	53.3	69.7	84.9	92.1	94.5		79.5	69.0	
Debt service of PPG external debt (in percent of exports + remittance	...	...	0.4			0.6	0.9	1.2	2.3	4.3	6.1		4.6	5.1	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2015–35

(Percent)

			Projections					
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to-GDP+remittances ratio								
Baseline	14	18	20	21	22	22	**20**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	14	13	13	14	15	16	**20**	25
A2. New public sector loans on less favorable terms in 2015-2035 2/	14	19	23	26	28	29	**31**	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	14	17	19	20	21	21	**19**	16
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	14	16	16	18	19	19	**18**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	14	17	19	21	21	21	**20**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	14	22	21	23	23	23	**21**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	14	9	-7	-4	-3	-1	**3**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	14	24	27	30	31	31	**29**	23
PV of debt-to-exports+remittances ratio								
Baseline	37	53	70	85	92	94	**79**	69
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	37	40	47	56	63	67	**78**	103
A2. New public sector loans on less favorable terms in 2015-2035 2/	37	57	80	104	118	125	**122**	125
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	37	51	68	83	90	92	**78**	68
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	37	45	53	66	73	75	**65**	60
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	37	51	68	83	90	92	**78**	68
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	37	68	73	89	96	98	**81**	69
B5. Combination of B1-B4 using one-half standard deviation shocks	37	26	-21	-15	-9	-5	**11**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	37	51	68	83	90	92	**78**	68
PV of debt-to-revenue ratio								
Baseline	64	86	84	90	90	90	**83**	61
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	64	64	56	60	62	64	**82**	91
A2. New public sector loans on less favorable terms in 2015-2035 2/	64	91	96	110	115	119	**128**	110
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	64	82	79	85	86	85	**79**	58
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	64	77	70	77	78	78	**75**	57
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	64	81	80	87	87	86	**81**	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	64	109	88	94	94	93	**85**	61
B5. Combination of B1-B4 using one-half standard deviation shocks	64	45	-29	-18	-10	-6	**13**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	64	118	116	126	126	125	**117**	85

Table A2. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt 2015–35 (Concluded)

(Percent)

Debt service-to-exports+remittances ratio								
Baseline	1	1	1	2	4	6	**5**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	1	1	2	3	5	**3**	5
A2. New public sector loans on less favorable terms in 2015-2035 2/	1	1	1	2	5	7	**4**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	6	**5**	5
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	1	1	2	4	5	**4**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	6	**5**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	1	2	2	4	6	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	0	2	3	**0**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	1	1	2	4	6	**5**	5
Debt service-to-revenue ratio								
Baseline	1	1	1	2	4	6	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	1	1	2	3	4	**3**	4
A2. New public sector loans on less favorable terms in 2015-2035 2/	1	1	1	3	5	7	**5**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	6	**5**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	1	1	2	4	6	**4**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	6	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	1	2	3	4	6	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	0	2	3	**0**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	2	2	3	6	8	**7**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	42	42	42	42	42	42	**42**	42

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A3. Liberia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–35

(Percent of GDP, fiscal year, unless otherwise indicated)

	Actual					Estimate						Projections			
	2012	2013	2014	Average 5/	Standard Deviation 5/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	11.0	12.4	16.1			24.6	29.8	33.7	36.6	38.3	39.2		35.0	27.2	
of which: foreign-currency denominated	11.0	12.4	16.1			24.6	29.8	33.7	36.6	38.3	39.2		35.0	27.2	
Change in public sector debt	-2.6	1.4	3.8			8.5	5.2	3.9	2.8	1.8	0.9		-1.1	-0.7	
Identified debt-creating flows	1.3	0.6	1.0			8.3	6.4	4.6	3.6	1.2	0.9		-1.7	-1.9	
Primary deficit	3.2	1.4	1.8	0.0	2.1	8.2	6.8	6.5	6.1	4.1	3.8	5.9	0.8	-0.1	0.8
Revenue and grants	27.9	29.9	27.4			32.5	32.1	30.3	27.1	27.5	27.6		26.6	29.0	
of which: grants	1.7	2.5	3.9			10.0	11.7	6.7	3.2	2.9	2.9		2.0	2.0	
Primary (noninterest) expenditure	31.2	31.4	29.2			40.6	38.9	36.8	33.2	31.6	31.4		27.4	29.0	
Automatic debt dynamics	-1.9	-0.8	-0.8			0.1	-0.4	-1.9	-2.4	-2.9	-2.9		-2.5	-1.8	
Contribution from interest rate/growth differential	-1.2	-0.9	-0.7			0.5	-0.9	-1.1	-1.7	-2.3	-2.5		-2.6	-1.3	
of which: contribution from average real interest rate	-0.2	0.0	0.0			0.0	0.0	-0.1	-0.3	-0.4	-0.4		-0.3	-0.2	
of which: contribution from real GDP growth	-1.0	-0.9	-0.7			0.5	-0.9	-1.0	-1.4	-1.9	-2.1		-2.4	-1.1	
Contribution from real exchange rate depreciation	-0.7	0.1	-0.1			-0.4	0.5	-0.8	-0.7	-0.6	-0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-3.9	0.8	2.8			0.2	-1.1	-0.7	-0.8	0.6	0.0		0.5	1.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	10.6			15.7	18.2	20.3	21.5	22.2	22.3		20.4	16.4	
of which: foreign-currency denominated	...	...	10.6			15.7	18.2	20.3	21.5	22.2	22.3		20.4	16.4	
of which: external	...	...	8.0			14.5	17.6	19.8	21.5	22.1	22.2		20.5	16.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.4	2.3	2.1			9.1	7.8	7.4	7.2	5.6	5.7		1.9	1.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	38.7			48.4	56.6	67.0	79.4	80.8	80.8		77.0	56.5	
PV of public sector debt-to-revenue ratio (in percent)	...	...	45.0			69.9	89.0	85.9	90.2	90.6	90.1		83.2	60.7	
of which: external 3/	...	...	34.2			64.5	85.9	83.7	90.0	90.3	89.9		83.5	60.6	
Debt service-to-revenue and grants ratio (in percent) 4/	0.4	3.0	1.1			2.8	3.2	3.0	4.1	5.4	6.8		4.2	4.2	
Debt service-to-revenue ratio (in percent) 4/	0.5	3.2	1.2			4.1	5.0	3.8	4.7	6.1	7.6		4.6	4.5	
Primary deficit that stabilizes the debt-to-GDP ratio	5.8	0.1	-2.0			-0.3	1.6	2.6	3.2	2.4	2.9		2.0	0.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.9	8.5	5.7	7.1	1.8	-2.8	3.9	3.6	4.5	5.5	5.8	3.4	7.1	4.2	6.1
Average nominal interest rate on forex debt (in percent)	0.2	1.4	1.3	0.6	0.6	1.4	1.1	1.0	0.9	0.8	0.9	1.0	1.2	1.2	1.2
Average real interest rate on domestic debt (in percent)	-5.7	-6.6	...	-5.8	4.4	...	...	...	...	...	...	...	...	...	...
Real exchange rate depreciation (in percent, + indicates depreciation)	-5.6	0.7	-0.9	-3.6	4.1	-2.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.7	8.0	9.2	9.5	3.5	7.4	6.5	9.9	7.8	8.3	8.6	8.1	4.7	7.0	6.0
Growth of real primary spending (deflated by GDP deflator, in percent)	24.5	9.1	-1.7	3.4	8.0	35.3	-0.5	-2.0	-5.8	0.5	5.1	5.4	6.5	3.3	5.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	46.9	53.2	52.1	52.7	53.9	53.9	52.1	44.6	44.6	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A4. Liberia: Sensitivity Analysis for Key Indicators of Public Debt, 2015–35

(Percent)

		Projections						
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	16	18	20	22	22	22	20	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	14	12	10	9	7	5	5
A2. Primary balance is unchanged from 2015	16	19	22	24	27	29	43	67
A3. Permanently lower GDP growth 1/	16	18	21	22	23	23	24	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	16	18	19	20	20	20	17	11
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	16	16	15	17	18	18	17	15
B3. Combination of B1-B2 using one half standard deviation shocks	16	15	13	14	15	14	11	5
B4. One-time 30 percent real depreciation in 2016	16	24	25	25	24	23	21	17
B5. 10 percent of GDP increase in other debt-creating flows in 2016	16	24	26	27	27	27	24	18
PV of Debt-to-Revenue Ratio 2/								
Baseline	48	57	67	79	81	81	77	57
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	48	44	41	38	33	27	18	17
A2. Primary balance is unchanged from 2015	48	59	73	90	99	107	163	231
A3. Permanently lower GDP growth 1/	48	57	68	81	84	85	90	94
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	48	56	64	74	74	73	63	36
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	48	48	50	62	64	65	64	50
B3. Combination of B1-B2 using one half standard deviation shocks	48	46	45	53	54	52	42	18
B4. One-time 30 percent real depreciation in 2016	48	75	82	91	88	84	79	59
B5. 10 percent of GDP increase in other debt-creating flows in 2016	48	74	85	98	99	98	91	63
Debt Service-to-Revenue Ratio 2/								
Baseline	3	3	3	4	5	7	4	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	3	3	3	3	4	5	2	1
A2. Primary balance is unchanged from 2015	3	3	3	4	6	7	6	11
A3. Permanently lower GDP growth 1/	3	3	3	4	6	7	5	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	3	3	3	4	5	6	4	3
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	3	3	3	3	5	6	3	4
B3. Combination of B1-B2 using one half standard deviation shocks	3	3	3	3	5	6	3	2
B4. One-time 30 percent real depreciation in 2016	3	3	4	5	7	9	7	7
B5. 10 percent of GDP increase in other debt-creating flows in 2016	3	3	3	5	6	7	5	5

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.